

Clustering in R

Kachun Huang

2025-04-21

Clustering in R

This tutorial introduces three common clustering methods in R: **Hierarchical Clustering**, **K-means Clustering**, and **Fuzzy C-Means Clustering**, using the `USArrests` dataset. It follows a standard structure: `load libraries` → `learn syntax` → `run examples`.

Load Required Libraries

```
# Core clustering and plotting libraries
library(stats)
library(cluster)
library(factoextra)
library(ggplot2)
```

Basic Syntax & Functions

Distance Matrix for Hierarchical Clustering

```
d <- dist(data)
```

- Computes pairwise **Euclidean distance** between observations

Hierarchical Clustering

```
hc <- hclust(d, method = "ward.D2")
```

Performs agglomerative clustering using:

- "ward.D2" – squared distances
 - other linkage strategies
 - * `ward.D` -Ward's minimum variance method (classic version)
 - * `single` -Minimum (single) linkage
 - * `complete` -Maximum (complete) linkage
 - * `average` -Average linkage (UPGMA)
 - * `centroid` -Centroid linkage (UPGMC)

```
# Dendrogram
plot(hc)

# Add rectangles for k clusters
rect.hclust(hc)
```

Plot Dendrogram or

```
# dendrogram from factoextra
fviz_dend(
  hc,
  k = 4, # num of clusters
)
```

K-means Clustering

```
fviz_nbclust(
  data,
  kmeans,
  method = "gap_stat"
)
```

Optimal Number of clusters Choose the k that maximizes the Gap statistic

```
km.res <- kmeans(data, centers = 2, nstart = 25)
```

- `centers`: number of clusters
- `nstart`: number of initial random configurations

```
fviz_cluster(km.res,
  data = data,
  ellipse.type = "convex",
  palette = "jco",
  ggtheme = theme_minimal()
)
```

Plot Clusters

Argument	Description
<code>fviz_cluster()</code>	Main function to plot clustering results from <code>kmeans()</code> , <code>hclust()</code> , or <code>fanny()</code>
<code>km.res</code>	The result object from <code>kmeans()</code> — contains cluster assignments and centroids

Argument	Description
<code>data = data</code>	The dataset that was clustered — used to calculate PCA for plotting
<code>ellipse.type = "convex"</code>	Draws convex hulls around each cluster to highlight boundaries (other options: <code>"norm"</code> , <code>"t"</code> for confidence ellipses)
<code>palette = "jco"</code>	Uses the “jco” color palette (from Journal of Clinical Oncology style) for clusters
<code>ggtheme = theme_minimal()</code>	Applies a clean and minimal ggplot2 theme to the plot for aesthetics

Fuzzy C-Means Clustering

K-means: Each point belongs to exactly one cluster (0/1)

Fuzzy C-Means: Each point belongs to all clusters to a degree (between 0 and 1)

```
res.fanny <- fanny(data, 2)
```

- Allows **soft membership** (each point belongs to multiple clusters with probabilities)

```
fviz_cluster(
  res.fanny,
  ellipse.type = "norm",
  repel = TRUE,
  palette = "jco",
  ggtheme = theme_minimal(),
  legend = "right"
)
```

Plot Clusters

Argument	Description
<code>fviz_cluster()</code>	Visualizes the clustering result, in this case from a fuzzy clustering object (<code>res.fanny</code>)
<code>res.fanny</code>	Clustering result from <code>fanny()</code> , which includes soft memberships for each point
<code>ellipse.type = "norm"</code>	Draws normal confidence ellipses (assuming multivariate normal distribution of points) around each cluster
<code>repel = TRUE</code>	Prevents text labels from overlapping by repelling them slightly apart for readability
<code>palette = "jco"</code>	Uses the “Journal of Clinical Oncology” (jco) color palette for cluster coloring
<code>ggtheme = theme_minimal()</code>	Applies a clean ggplot2 theme with minimal gridlines and distractions
<code>legend = "right"</code>	Places the cluster legend on the right side of the plot for clarity

Example

Import Data

```
data <- USArrests
data <- na.omit(data)
data <- scale(data)
data <- as.data.frame(data) # Ensure it's a data frame
```

Hierarchical Clustering

Distance Matrix for Hierarchical Clustering

```
d <- dist(data)
```

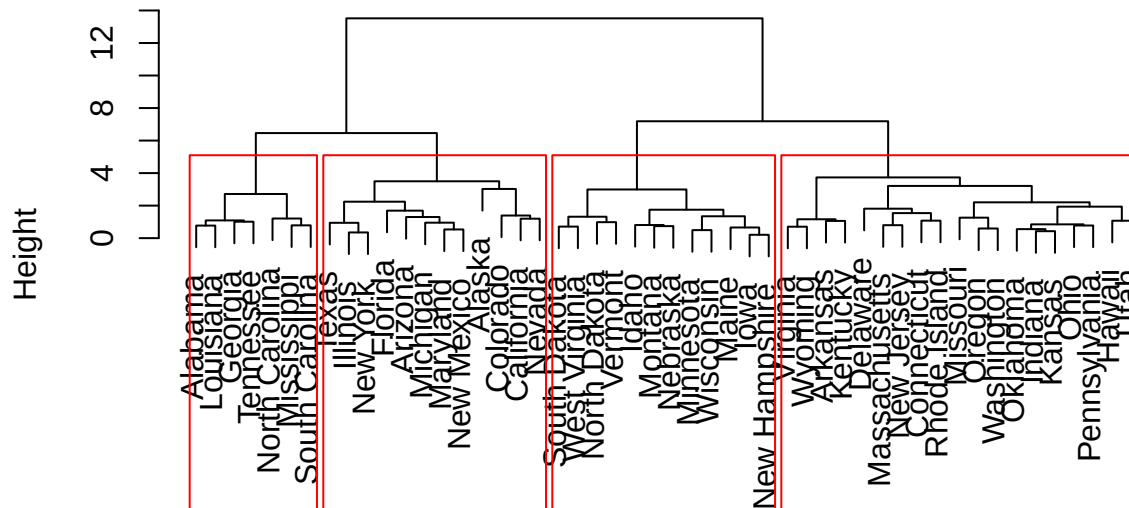
Hierarchical Clustering

```
hc <- hclust(d, method = "ward.D2")
```

Plot

```
plot(hc, main = "Hierarchical Clustering Dendrogram")
rect.hclust(hc, k = 4, border = "red")
```

Hierarchical Clustering Dendrogram



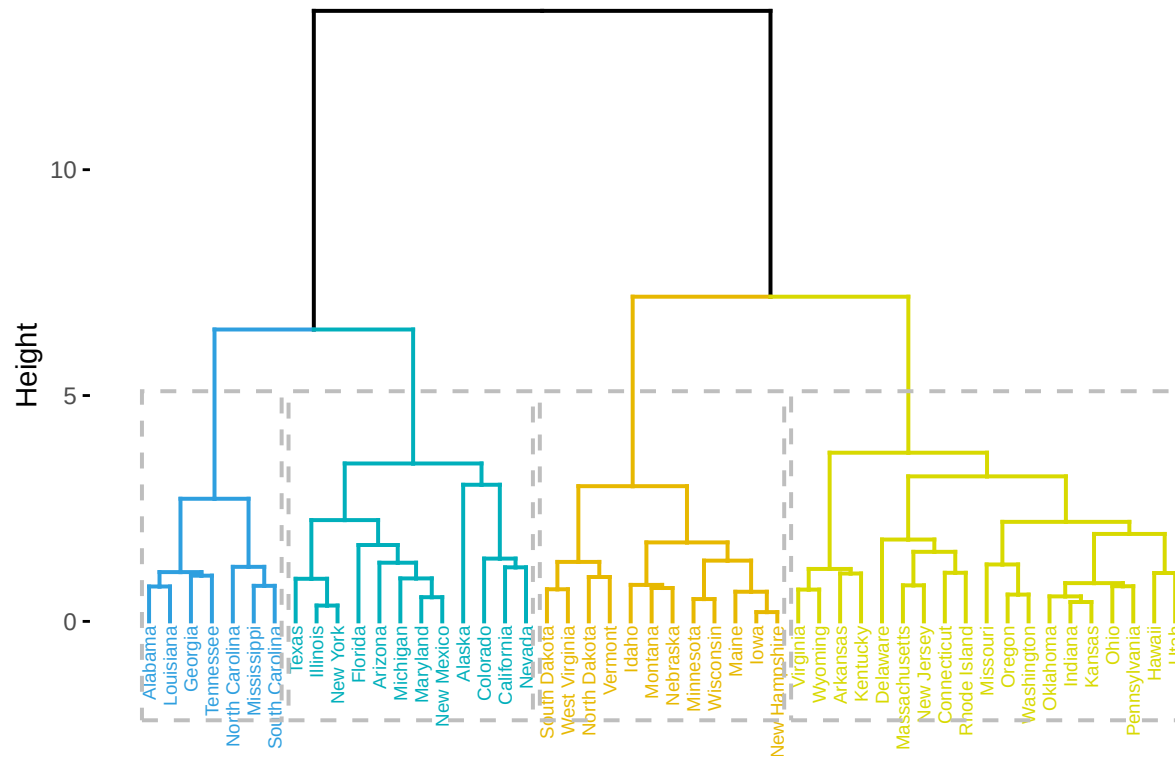
d
hclust (*, "ward.D2")

or

```
fviz_dend(
  hc,
  k = 4,
  cex = 0.5, # Label size
  k_colors = c("#2E9FDF", "#00AFBB", "#E7B800", "#D8D900"),
  color_labels_by_k = TRUE, # Color labels by cluster
  rect = TRUE # Draw rectangles around clusters
)
```

```
## Warning: The '<scale>' argument of 'guides()' cannot be 'FALSE'. Use "none" instead as
## of ggplot2 3.3.4.
## i The deprecated feature was likely used in the factoextra package.
## Please report the issue at <https://github.com/kassambara/factoextra/issues>.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

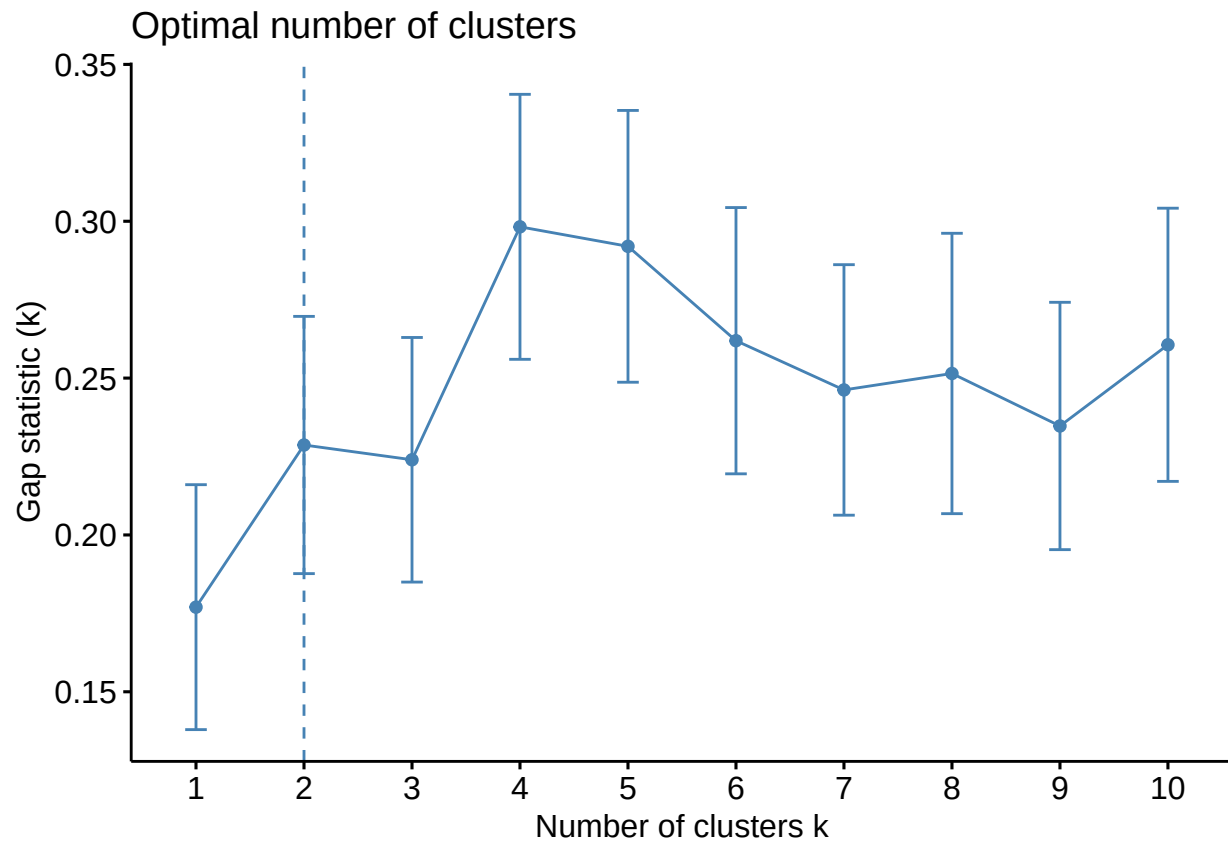
Cluster Dendrogram



K-means Clustering

Optimal Number of clusters

```
fviz_nbclust(data, kmeans, method = "gap_stat")
```

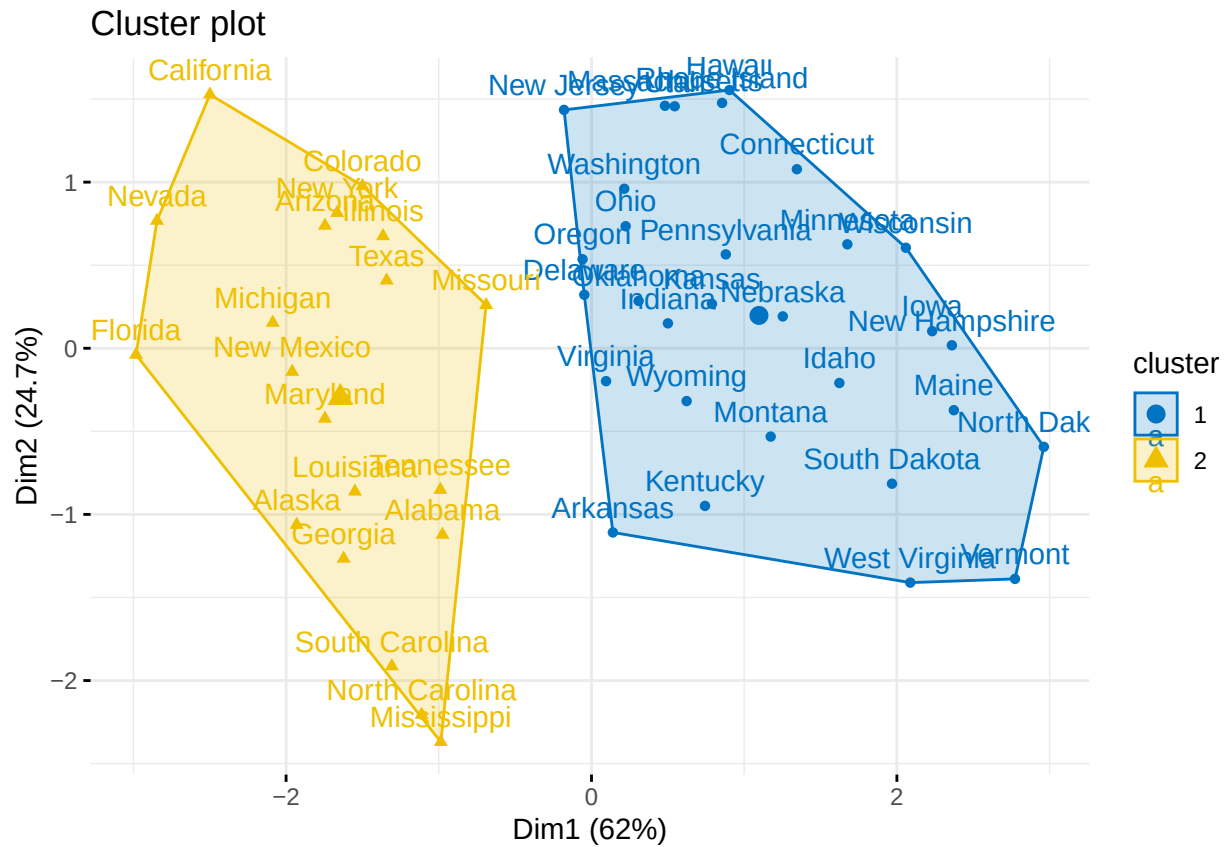


K-means Clustering

```
km.res <- kmeans(data, centers = 2, nstart = 25)
```

Plot

```
fviz_cluster(km.res,  
  data = data,  
  ellipse.type = "convex",  
  palette = "jco",  
  ggtheme = theme_minimal()  
)
```



Fuzzy C-means Clustering

Fitting model and Plot

```
res.fanny <- fanny(data, 2)

fviz_cluster(
  res.fanny,
  ellipse.type = "norm",
  repel = TRUE,
  palette = "jco",
  ggtheme = theme_minimal(),
  legend = "right"
)
```

Cluster plot

